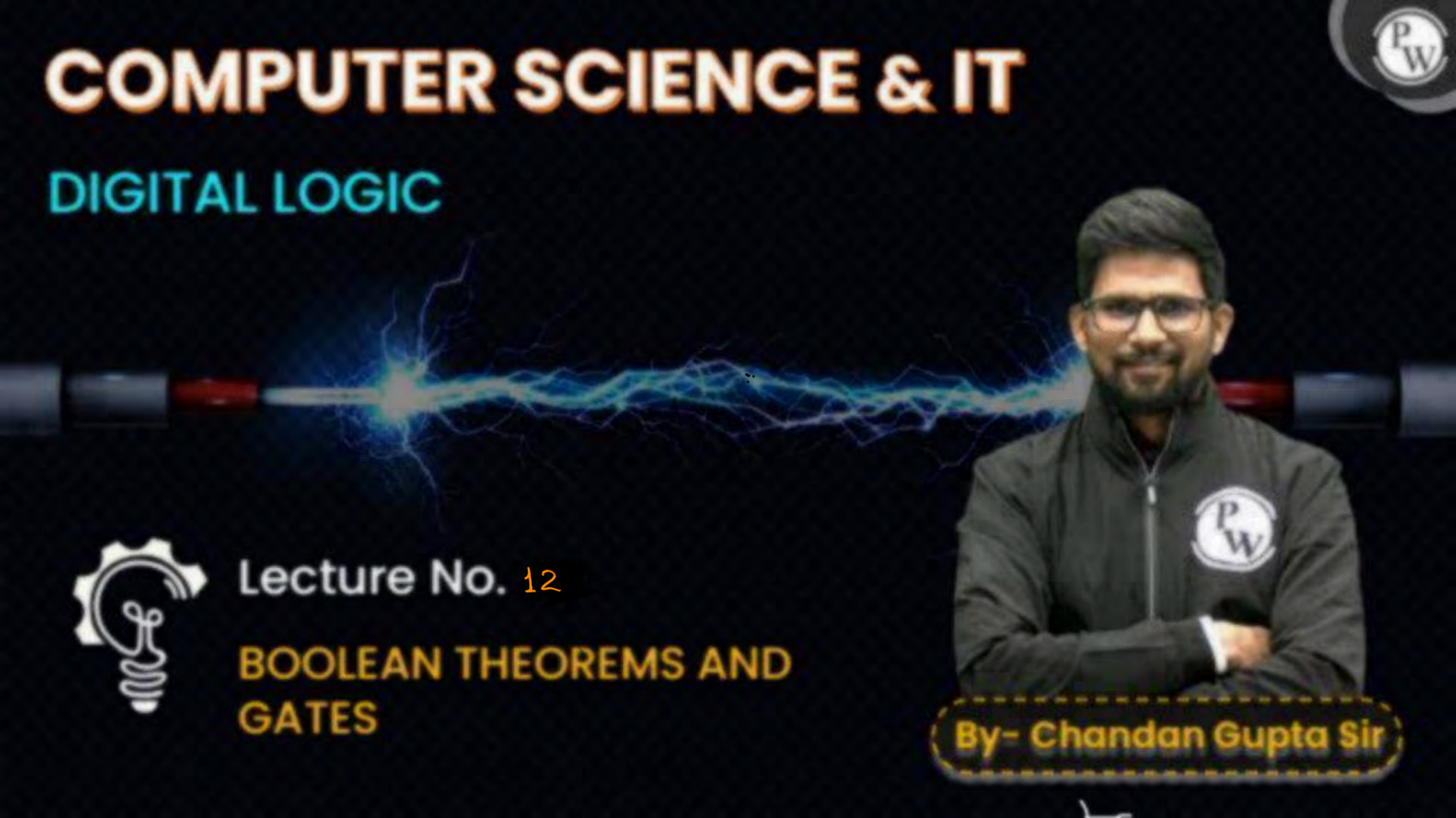
# COMPUTER SCIENCE & IT

## DIGITAL LOGIC



Lecture No. 12

BOOLEAN THEOREMS AND GATES

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9mp basics.




Question Discussion

H.W

Q.1. $f(A,B,C,D) = \Sigma(0,1,2,3,7,9,10,11)$ Self dual or NOT

Q.2. $f(A,B,C) = \Sigma(1,2,3,6)$ Self dual or NOT

Q.3 *** $f(A,B,C) = AB \oplus BC \oplus AC \longrightarrow$ self dual or NOT

Q.4. $f(A,B,C,D) = \Sigma(1,2,3,7,9,10,11,15)$ Self dual or NOT

Q.5 *** $f(A,B,C) \longrightarrow$ is self dual function

then $\overline{f}(A,B,C)$ will be self dual or NOT ?

Answer Keys:

1. Self dual
2. Non-self dual
3. Self dual
4. Self dual
5. Yes.

- $AB \oplus BC \oplus AC = f(A,B,C) = \Sigma(3,5,6,7) = AB + BC + CA$

$\Sigma(6,7) \oplus \Sigma 3,7 \quad \oplus \quad \Sigma(5,7)$

$f_1(A,B,C) \oplus f_2(A,B,C) \oplus f_3(A,B,C) =$

$\Rightarrow A \cdot B$

$1\ 1\ 0 \rightarrow 6 \quad AB\bar{C}$

$1\ 1\ 1 \rightarrow 7 \quad ABC$

$f_1(A,B,C) = \Sigma(6,7)$

$BC \rightarrow \Sigma(3,7)$

$0\ 1\ 1$

$1\ 1\ 1$

$A\ C$

$1\ 0\ 1 \rightarrow 5$

$1\ 1\ 1 \rightarrow 7$

$AB \oplus BC = B[A \oplus C]$

$\Sigma(6,7) \oplus \Sigma(3,7) = \Sigma(3,6)$

$= \bar{A}BC + AB\bar{C}$

$= B[\bar{A}C + A\bar{C}]$

$= B(A \oplus C)$

$$A\underline{B} \oplus \underline{\bar{B}}C = AB + \bar{B}C$$

$$\Sigma(6,7) \quad \Sigma(1,5) = \Sigma(1,5,6,7)$$

$$= \bar{B}C + AB$$

| $\bar{B}$ | $C$ | | |
|---|---|---|---|
| 0 | 0 | 1 | → 1 |
| 1 | 0 | 1 | → 5 |

$$\bar{A}B \oplus A\bar{B} = \bar{A}B + A\bar{B}$$

$$= A \oplus B$$

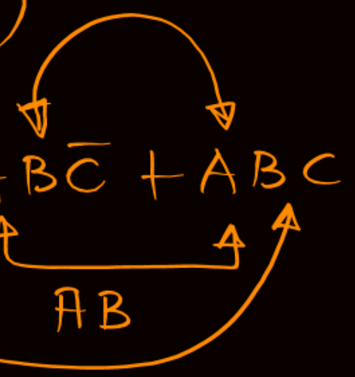
$$\bar{A}\bar{B} \oplus AB = \bar{A}\bar{B} + AB$$

$$= A \odot B$$

$$AB \oplus BC \oplus ABC = AB + BC$$

$$\Sigma(6,7) \quad \Sigma(3,7) \quad \Sigma 7 = \Sigma(3,6,7)$$

$$= \bar{A}BC + AB\bar{C} + ABC$$




$$= AB + \bar{A}BC$$

$$= B\left[A + \bar{A}C\right]$$

$$= B\left[A + C\right]$$

$$= AB + BC.$$

# [MCQ]

The dual of a Boolean function $F(x_1, x_2, ..., x, +, \cdot, ')$, written as $F^D$, is the same expression as that of F with + and · swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

[GATE-2022-CS: 1M]

$n \longrightarrow N = 2^n \longrightarrow$ Comb.

$\longrightarrow M = 2^N$

$S_D = 2^{N/2} = 2^{2^{n-1}}$

A. 2n

B. $2^{n-1}$

C. $2^{2^n}$

D. $2^{2^{n-1}}$

# [MCQ]

If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

[GATE-2017-CS: 1M]

A. $wx + w(x + y) + x(x + y) = x + wy$

B. $\overline{w\bar{x}(y + \bar{z})} + \bar{w}x = \bar{w} + x + \bar{y}z$

C. $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$

D. $(w + y)(wxy + wyz) = wxy + wyz$

$x + wx + wy$

$x + wy$

$\bar{w} + x + \overline{y+z} + \bar{w}x$

$\bar{w} + x + \bar{y}\cdot z$

$\bar{x}y$

$[w\bar{x}y + \bar{w}\bar{x}]y$

$= \bar{x}y[wy + \bar{w}] = \bar{x}y\cdot[\bar{w} + y]$

$= \bar{x}[y]$

$\bar{w} + (wy)$

$\Rightarrow w(w+y)(xy + yz)$

$= w(xy + yz)$

$= wxy + wyz$

# [MCQ]

The binary operator ≠ is defined by the following truth table

| P | Q | p≠q |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$P \oplus Q$

$\neq \; - \; \oplus$

Which one of the following is true about the binary operator ≠?

[GATE-2015-CS: 1M]

A. Both commutative and associative

B. Commutative but not associative

C. Not commutative but associative

D. Neither commutative nor associative

# [MCQ]

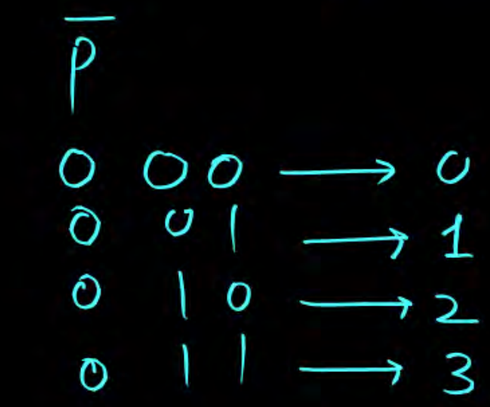
Given the function F = P' + QR, wehre F is a function in three Boolean variables P,Q and R and P' = !P, consider the following statements.

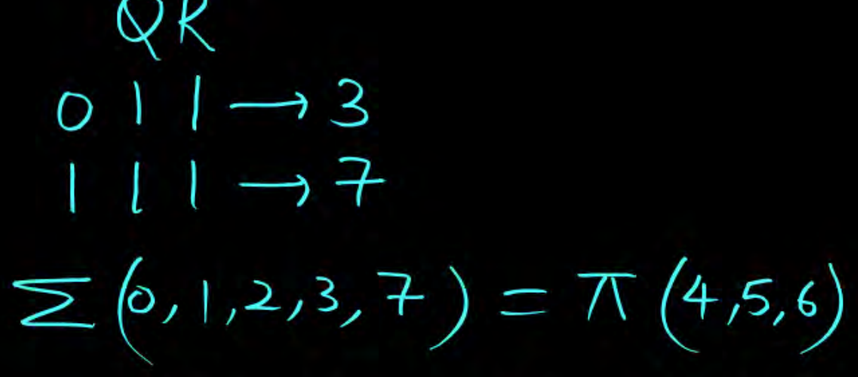
(S1) $F = \Sigma(4, 5, 6)$

(S2) $F = \Sigma(0,1,2,3,7)$

(S3) $F = \Pi(4, 5, 6)$

(S4) $F = \Pi(0,1,2,3,7)$




[GATE-2015-CS: 2M]

Which of the following is true?

A. (S1) – False, (S2) – True, (S3) – True, (S4) -False

B. (S1) – True, (S2) – False, (S3) – False, (S4) -True

C. (S1) – False, (S2) – False, (S3) – True, (S4) – True

D. (S1) – True, (S2) – True, (S3) – False, (S4) -False

# [MCQ]

The truth table

| X | Y | F(X, Y) |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

$F(x,y) = \sum(2,3)$

$= x\bar{y} + xy$

$= x(1) = x$

represents the Boolean function

[GATE-2012-CS: 1M]

A) x

B) x+y

C) $x \oplus y$

D) y

# [MCQ]

The simplified SOP (sum of product) form of the Boolean expression

$(P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + R) \cdot (P + Q + \bar{R})$

[GATE-2011-CS: 1M]

$$\left[ (P+\bar{Q}) + \bar{R} \cdot R \right] \left[ P+Q+\bar{R} \right] = (P+\bar{Q})(P+Q+\bar{R})$$

$$= P + \bar{Q} \cdot (Q+\bar{R})$$

$$= P + \bar{Q} \cdot \bar{R}$$

A. $(\bar{P}.Q + \bar{R})$

B. (P + Q .'R')

C. (P'.Q + R)

D. (P.Q + R)

# [MCQ]

The minterm expansion of $f(P, Q, R) = PQ + Q\overline{R} + P\overline{R}$ is

[GATE-2011-CS: 1M]

$\Sigma(2,4,6,7)$

$= \pi(0,1,3,5)$

Self dual.

PQ

1 1 0 → 6

1 1 1 → 7

SOP → minterm m

POS → Max term M

$Q\overline{R}$

0 1 0 → 2

1 1 0 → 6

$P\overline{R}$

1 0 0 → 4

1 1 0 → 6

$M_0 \cdot M_1 \cdot M_3 \cdot M_5$

A. $m_2 + m_4 + m_6 + m_7$

B. $m_0 + m_1 + m_3 + m_5$

C. $m_0 + m_1 + m_6 + m_7$

D. $m_2 + m_3 + m_4 + m_5$

$M_0 + M_1 + M_3 + M_5$

# [MCQ]

If P, Q, R are Boolean variables, then $(P + Q')\ (P.Q' + P.R)\ (P'.R' + Q')$ Simplifies to

[GATE-2008-CS: 1M]

A. $P.\overline{Q}$

B. $P.\overline{R}$

C. $P.\overline{Q}+R$

D. $P.\overline{\overline{R}}+Q$

$$= \left[\overline{Q} + P.\overline{P}.\overline{R}\right]\left[P\overline{Q} + PR\right]$$

$$= \overline{Q}P\left[\overline{Q} + R\right]$$

$$= P\,\overline{Q}.\left[\overline{Q} + R\right]$$

$$= P.\overline{Q}$$

$$(P+\overline{Q})\ P\ (\overline{Q}+R)\ (\overline{P}\overline{R}+\overline{Q})$$

$$P\ (\overline{Q}+R)\ (\overline{P}\overline{R}+\overline{Q})$$

$$P\left[\overline{Q} + R.\overline{P}\overline{R}\right.$$

[MSQ]

Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where $x_1, x_2, x_3, x_4$ are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be TRUE?

[GATE-2022-CS: 1M]

$x_1, x_2, x_3, x_4$

0, 1, 2, 3, 4

A. $x_1 x_2 x_3 x_4 = 0$

B. $x_1 x_3 + x_2 x_4 = 0$

C. $\bar{x_1} \oplus \bar{x_3} = \bar{x_2} \oplus \bar{x_4}$

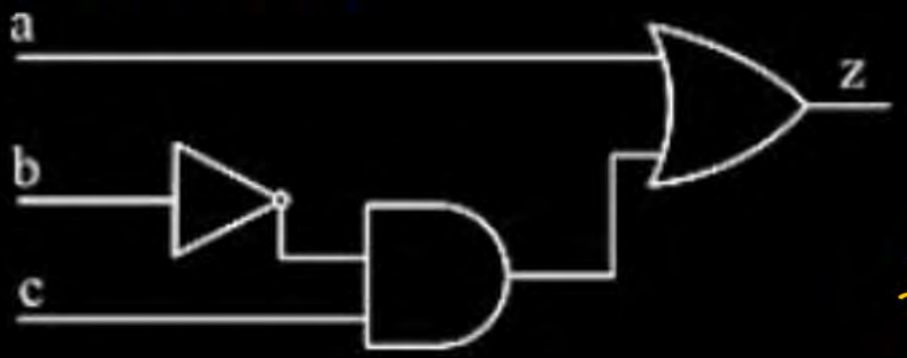
D. $x_1 + x_2 + x_3 + x_4 = 0$

$x_1 \oplus x_3 = x_2 \oplus x_4$

# [MCQ]

Consider the Boolean function z(a,b,c). Which one of the following minterm lists represents the circuit given above?

[GATE-2020-CS: 2M]




ft. W.

A $Z = \Sigma (0, 1, 3, 7)$

B $Z = \Sigma (2, 4, 5, 6, 7)$

C $Z = \Sigma (1, 4, 5, 6, 7)$

D $Z = \Sigma (2, 3, 5)$

# [MCQ]

Which one of the following is NOT a valid identity?

[GATE-2019-CS: 1M]

A. $(x + y) \oplus z = x \oplus (y + z)$

B. $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

C. $x \oplus y = x + y$, if $xy = 0$

D. $x \oplus y = (xy + x'y')'$

$x \odot y$

$(x+y) \oplus z \neq x \oplus (y+z)$ ×

$x \oplus y \odot z = x \odot y \oplus z$

$x \oplus y = (x+y)$ if $x \cdot y = 0$

| | |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |

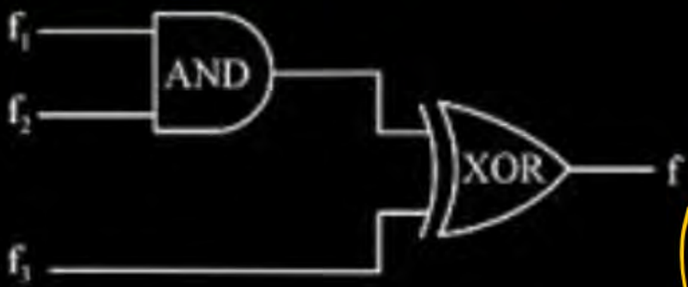
# [MCQ]

Consider three 4-variable functions f1, f2 and f3, which are expressed in sum-of-minterms as

$f_1 = \Sigma(0, 2, 5, 8, 14)$, $f_2 = \Sigma(2, 3, 6, 8, 14, 15)$, $f_3 = \Sigma(2, 7, 11, 14)$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:

[GATE-2019-CS: 2M]




(H.W).

A. $\Sigma(2, 14)$

B. $\Sigma(7, 8, 11)$

C. $\Sigma(2, 7, 8, 11, 14)$

D. $\Sigma(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

# [MCQ]

Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT? [GATE-2018-CS: 1M]

A. $P \oplus Q = \overline{P \odot Q}$

B. $\overline{P} \oplus Q = P \odot Q$

C. $P \oplus \overline{Q} = \overline{P} \oplus Q$

D. $(P \oplus \overline{P}) \oplus Q = (P \odot \overline{P}) \odot \overline{Q}$

# [MCQ]

Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q:

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

[GATE-2014-CS: 1M]

A. P + Q

B. (P + Q)'

C. P ⊕ Q

D. (P ⊕ Q)'

$P \oplus P \oplus P = P$

$0 \oplus P = P$

$P \oplus P \oplus P \oplus P \oplus P = P$

$P \oplus \overline{P} \oplus P =$

$1 \oplus P = \overline{P}$

$Q \oplus Q \oplus Q = Q$

$P \oplus Q \oplus 1 = P \oplus \overline{Q}$

$= P \odot Q$

$= \overline{P \oplus Q}$

# [MCQ]

Which one of the following circuits is NOT equivalent to a 2-input X-NOR (exclusive NOR) gate?

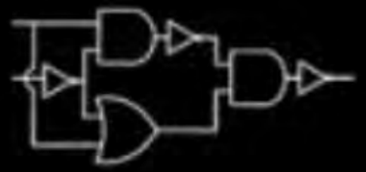
[GATE-2011-CS: 1M]

H.W

A 

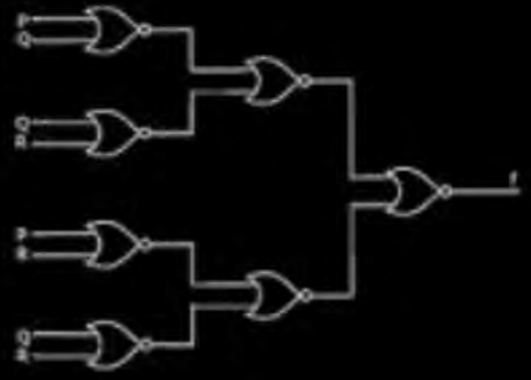
B 

C 

D 

# [MCQ]

What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below? [GATE-2010-CS: 1M]



A $\overline{Q+R}$

B $\overline{P+Q}$

C $\overline{P+R}$

D $\overline{P+Q+R}$

# [MCQ]

What is the minimum number of gates required to implement the Boolean function (AB + C) if we have to use only 2-input NOR gates? [GATE-2009-CS: 1M]

H.W

A 2

B 3

C 4

D 5

# [MCQ]

Given f1,f3 and f in canonical sum of products from (in decimal) for the circuit.

[GATE-2008-CS: 1M]




$f = f_4 + f_3$

$F_1 = \sum m\,(4,5,6,7,8)$

$F_3 = \sum m\,(1, 6, 15)$

$F = \sum m\,(1, 6, 8, 15)$

Then $f_2$ is

(A) $\sum m\,(4, 6)$

(B) $\sum m\,(4, 8)$

(C) $\sum m\,(6, 8)$

(D) $\sum m\,(4, 6, 8)$

$f_4 \rightarrow \Sigma$ ( terms in both $f_1$ & $f_2$ )

$f_2 \rightarrow$ 4X, 5X, 7X

$f_2 \rightarrow$ 6, 8 $\checkmark$ (0, 1, 2, 3, 9, --- 15)

8

$\sum_{2^{11}}(6,8)$

(8) - $2^{11}$

$^{11}C_0 + {}^{11}C_1 - - - {}^{11}C_{11} = 2^{11}$

$2^{11} + 2^{11} = 2^{12}$

(1, 8)

(0, 6, 8)

## Question

A logical function f(A, B, C) is given as $f(A,B,C) = \overline{A \oplus B \oplus C} + (A \oplus B)\overline{C}$

Then minimum no. of 2-input NAND gate required to implement f(A, B, C) will be _____.



# 2 Minute Summary

Question Practice



# Happy Learning

# THANK - YOU